

Project Proposal

Exploring the impact of visible homelessness in Retail Store Performance.



Our Team



Ricardo Harrisson



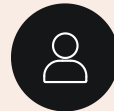
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The Core Question



How does homelessness in downtown Toronto affect local retail sales and foot traffic patterns?



The Downtown Toronto Context

Pinpointing the geographic and socio-economic specifics of a major metropolitan area.



Impact on Local Retail Sales

Assessing how homelessness influences economic activity and business performance.



Shifts in Foot Traffic Patterns

Analyzing changes in pedestrian movement and public space utilization.

Our Data Sources

Unlocking Insights from City of Toronto Data



Three Datasets:

- Toronto Economic Dashboard – Retail Sales (in \$ Billions)
- Daily Shelter Overnight Service Occupancy & Capacity
- Traffic Volumes - Multimodal Intersection Turning Movement Counts



Sources

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 City of Toronto

Toronto's Retail Sales Dataset

The Toronto Economic Dashboard offers a timely bird's-eye view of Toronto's econom...

 City of Toronto Open Data Portal

Toronto Homelessness Shelter Dataset

 City of Toronto Open Data Portal

Toronto Foot Traffic Dataset



Dataset Snapshot

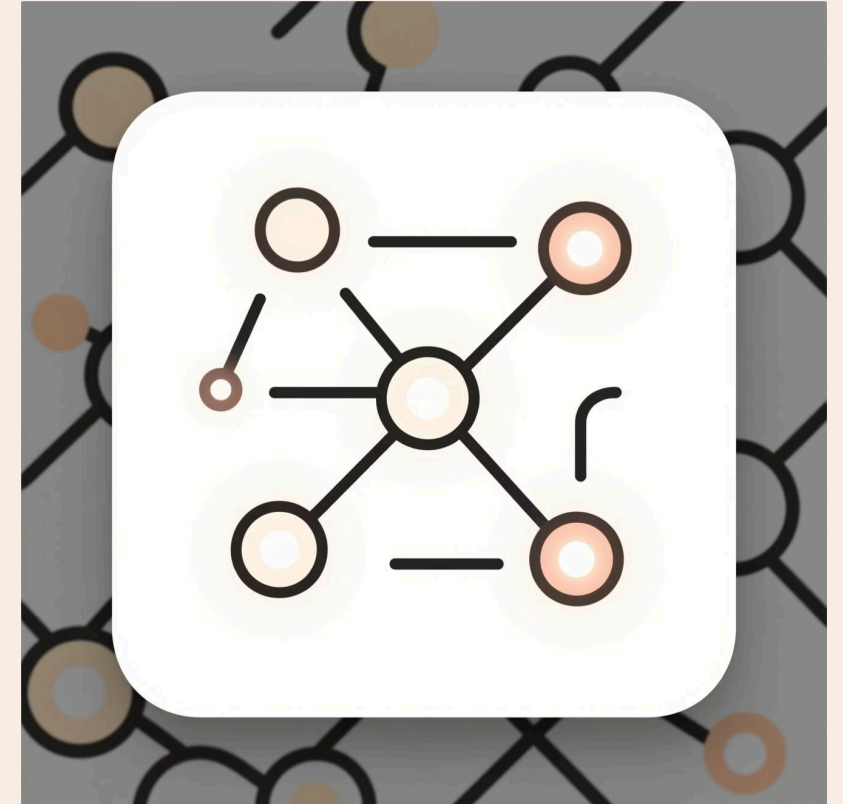
Our datasets provides a detailed look at **shelter usage**.

Combined datasets for year 2021 - 2022 data

- 100,422 observations
- 32 variables

Key variables for our analysis:

- **Location:** Where shelters are situated.
- **Capacity:** Available beds/spaces.
- **Service User Count:** Daily occupancy figures.





Dataset Snapshot

The datasets to follow provide comprehensive views of **retail activity** and **urban mobility**.

Retail Sales

Track economic trends and business health.

- 80 observations
- 2 variables

Traffic Volumes

Analyze pedestrian and vehicular movement patterns.

- 6,312 observations
- 34 variables

These rich datasets will allow us to correlate different urban dynamics and identify patterns.

Similar Initiatives

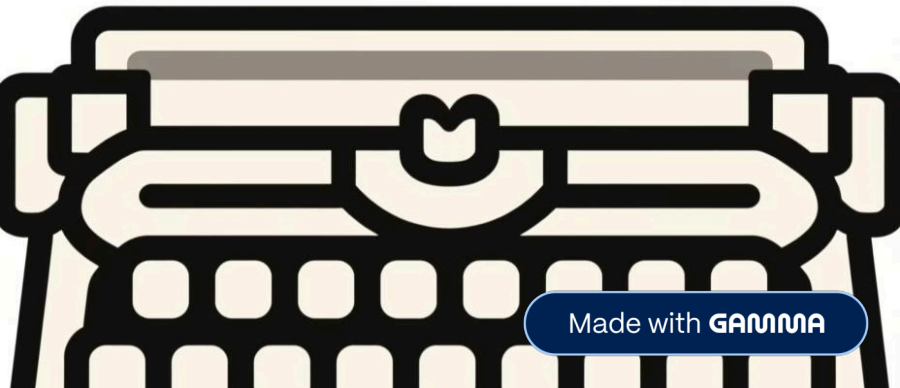
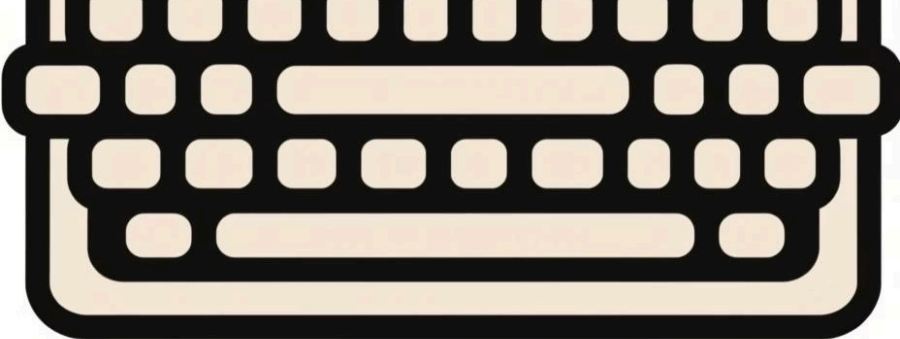
Two key projects inform our approach:

Shelter System Flow

City of Toronto

Downtown Recovery Dashboard

University of Toronto



Shelter System Flow: Insights

City of Toronto's monthly data on shelter usage.

Active Homelessness

People using shelters in the last 3 months, not in permanent housing.

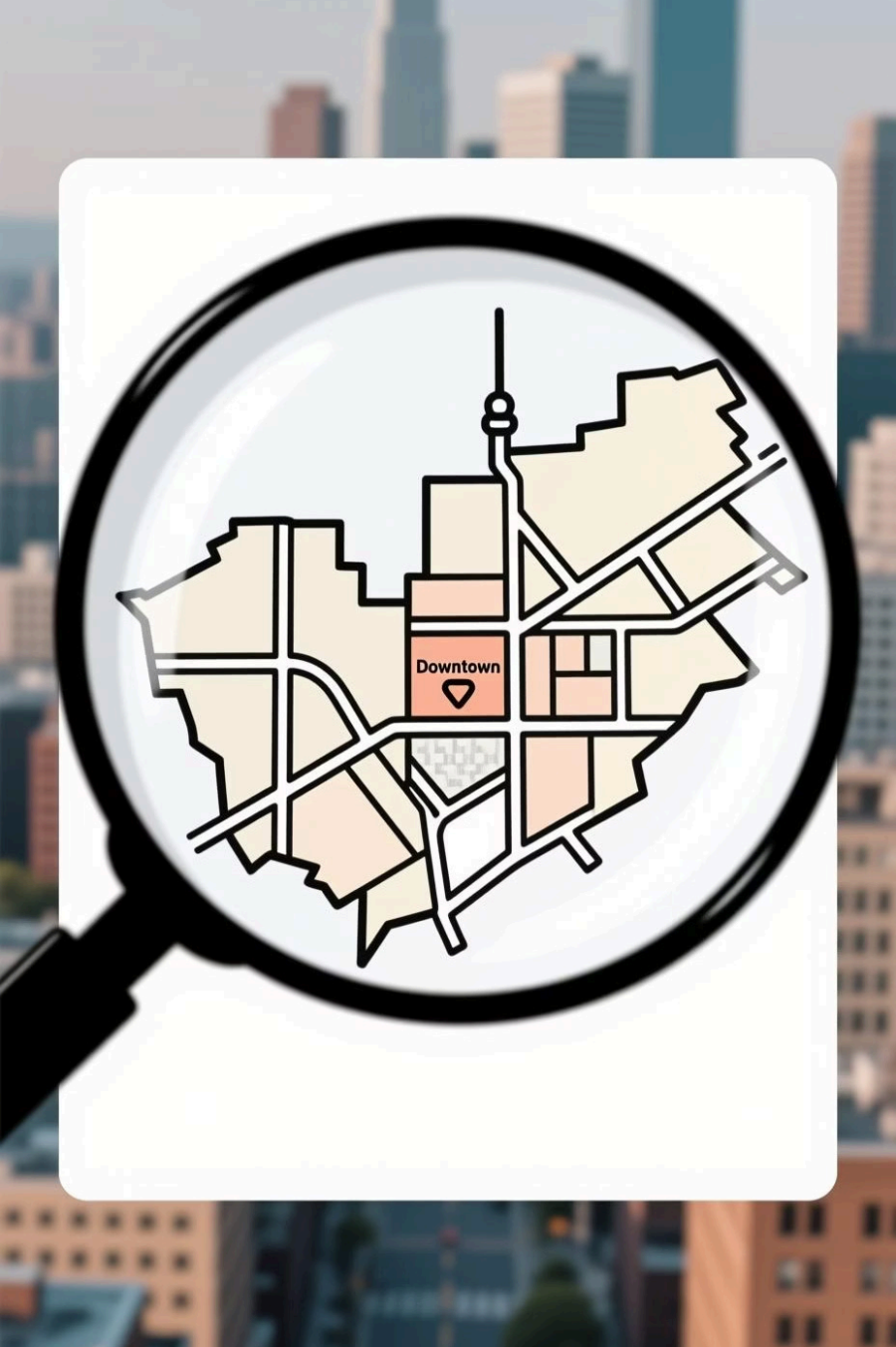
Key Metrics

Inflow, outflow, and active homeless population.

Subpopulation Data

Disaggregated by families, single adults, youth, chronically homeless.





Shelter System Flow: Limitations

While valuable, this data has constraints for our project.



Monthly Totals Only

Lacks daily granularity for short-term trends.



City-Wide Scope

No neighborhood-level precision.



No Economic Link

Focuses solely on homelessness, not economic indicators.

Downtown Recovery Dashboard

University of Toronto's broad view of urban vitality.



Evaluates Vitality

Uses anonymized mobile data to track unique visitors.



Tracks Recovery

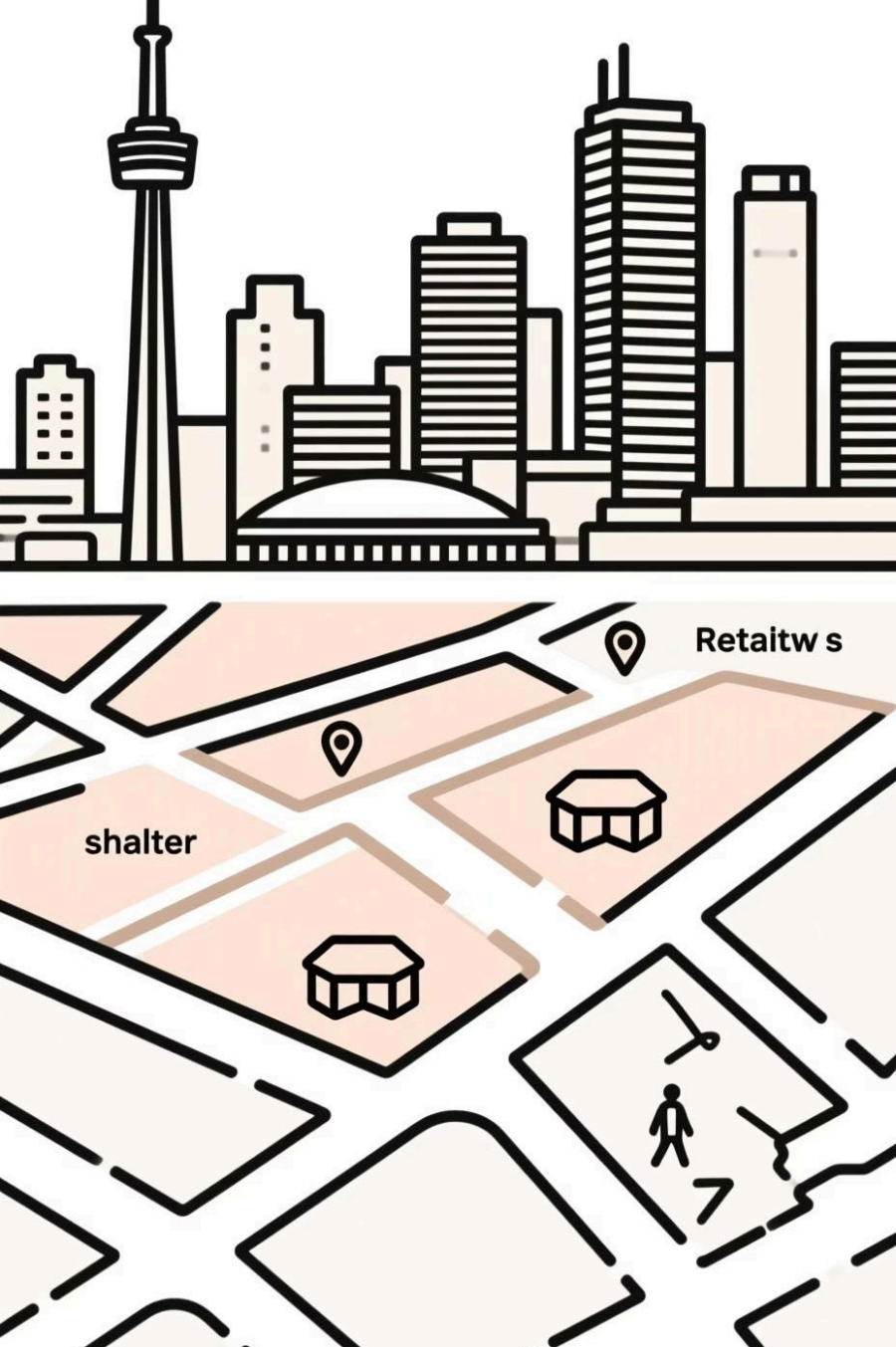
Assesses post-pandemic recovery across multiple cities.



Broad Perspective

Useful for big-picture overview, but less granular.





Our Unique Contribution

Bridging the gap between social factors and economic outcomes.



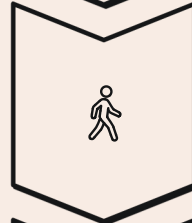
Daily Shelter Data

Granular, localized insights.



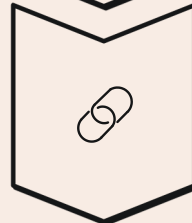
Retail Activity

Direct economic impact.



Foot Traffic

Movement patterns.



Predictive Relationships

Connecting homelessness to downtown activity.

Results



Results (Continued)



Linear Regression: Does the model work?

Model explains 19% of retail sales variation ($R^2 = 0.192$) → weak fit

Overall model is not statistically significant ($p = 0.278$)

Homelessness:

Coefficient is positive but not meaningful

No statistically significant relationship with retail sales ($p = 0.486$)

Foot Traffic:

Shows a weak positive trend

Still not statistically significant ($p = 0.150$)

Not a strong predictor of retail performance

Interpretation:

Homelessness and foot traffic do not reliably explain changes in Toronto retail sales

Retail likely driven by other factors (inflation, seasonality, economic activity, events)

[14]:

OLS Regression Results							
Dep. Variable:	retail_dollars			R-squared:	0.192		
Model:	OLS			Adj. R-squared:	0.058		
Method:	Least Squares			F-statistic:	1.428		
Date:	Tue, 18 Nov 2025			Prob (F-statistic):	0.278		
Time:	15:19:48			Log-Likelihood:	-327.38		
No. Observations:	15			AIC:	660.8		
Df Residuals:	12			BIC:	662.9		
Df Model:	2						
Covariance Type:	nonrobust						
	coef	std err	t	P> t	[0.025	0.975]	
Intercept	6.132e+09	5.79e+09	1.060	0.310	-6.48e+09	1.87e+10	
homeless_count	1.487e+06	2.07e+06	0.719	0.486	-3.02e+06	6e+06	
pedestrian_count	5.746e+04	3.73e+04	1.539	0.150	-2.39e+04	1.39e+05	
Omnibus:	1.351	Durbin-Watson:	0.742				
Prob(Omnibus):	0.509	Jarque-Bera (JB):	0.951				
Skew:	-0.580	Prob(JB):	0.622				
Kurtosis:	2.581	Cond. No.	2.65e+05				

Cross Validation:

Model Fit & Stability

Model trained on 2021 data shows moderate error, but performance drops sharply on 2022 data.

Large increase in validation error indicates the model does not generalize well to new data.

Performance Metrics:

Train RMSE (2021): ~447M

Validation RMSE (2022): ~1.94B

Validation error is 4× higher, showing the model struggles to make accurate predictions outside the training period.

Predictor Significance:

Neither homelessness nor pedestrian traffic is a statistically meaningful predictor of retail sales.

Coefficients are unstable and not significant across train/validation splits.

Interpretation:

The model is likely overfitting to 2021 patterns that do not repeat in 2022.

Retail performance appears driven by other economic factors rather than homelessness or foot-traffic counts.

Train rows (2021): 6
Validation rows (2022): 9

OLS Regression Results

Dep. Variable:	retail_dollars	R-squared:	0.570
Model:	OLS	Adj. R-squared:	0.283
Method:	Least Squares	F-statistic:	1.987
Date:	Tue, 18 Nov 2025	Prob (F-statistic):	0.282
Time:	15:20:03	Log-Likelihood:	-128.03
No. Observations:	6	AIC:	262.1
Df Residuals:	3	BIC:	261.4
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-7.422e+09	1.03e+10	-0.723	0.522	-4.01e+10	2.53e+10
homeless_count	6.344e+06	3.88e+06	1.634	0.201	-6.01e+06	1.87e+07
pedestrian_count	-7.551e+04	1.46e+05	-0.518	0.640	-5.4e+05	3.89e+05

Omnibus:	nan	Durbin-Watson:	2.193
Prob(Omnibus):	nan	Jarque-Bera (JB):	0.429
Skew:	-0.206	Prob(JB):	0.807
Kurtosis:	1.757	Cond. No.	2.59e+05

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly spe

[2] The condition number is large, 2.59e+05. This might indicate that there are strong multicollinearity or other numerical problems.

Model Performance:
Train RMSE (2021): 447,364,524.36
Validation RMSE (2022): 1,945,967,182.81